

MFE5200 – Computational Methods in Financial Engineering
Summer Term, 2025-26
Assignment 3 (Solution)

Problem 1

Let $\hat{X}_1$ and $\hat{X}_2$ be the time-discretized approximation of X_1 and X_2 respectively.

Let $Z_1(t)$ and $Z_2(t)$ be two independent standard Brownian motions.

Since $W_1(t)$ and $W_2(t)$ are two correlated Brownian motions and $dW_1(t)dW_2(t) = \rho dt$, we have

$$\begin{aligned}dW_1(t) &= dZ_1(t), \\dW_2(t) &= \rho dZ_1(t) + \sqrt{1 - \rho^2} dZ_2(t).\end{aligned}$$

$$\hat{X}_1(0) = X_1(0) \text{ and } \hat{X}_2(0) = X_2(0).$$

For $i = 0, 1, \dots, m - 1$,

$$\begin{aligned}\hat{X}_1(t_{i+1}) &= \hat{X}_1(t_i) + a_1 \hat{X}_1(t_i)(t_{i+1} - t_i) + \sigma_1 \hat{X}_1(t_i) \sqrt{t_{i+1} - t_i} R_{i+1}, \\ \hat{X}_2(t_{i+1}) &= \hat{X}_2(t_i) + a_2 \hat{X}_2(t_i)(t_{i+1} - t_i) + \sigma_2 \hat{X}_2(t_i) \sqrt{t_{i+1} - t_i} \left(\rho R_{i+1} + \sqrt{1 - \rho^2} V_{i+1} \right).\end{aligned}$$

Problem 2

Let $Z_1, Z_2, \dots, Z_m$ be independent standard normal random variables.

Let $\hat{X}$ be the time-discretized approximation of X_1 and X_2 respectively.

Euler scheme:

$$\hat{X}(0) = X_0.$$

For $i = 0, 1, \dots, m - 1$,

$$\hat{X}(t_{i+1}) = \hat{X}(t_i) + r \hat{X}(t_i)(t_{i+1} - t_i) + \sqrt{\sigma_0^2 + \sigma_1^2 \hat{X}^2(t_i)} \sqrt{t_{i+1} - t_i} Z_{i+1}.$$

Milstein scheme:

$$\hat{X}(0) = X_0.$$

For $i = 0, 1, \dots, m-1$,

$$\begin{aligned}
\hat{X}(t_{i+1}) &= \hat{X}(t_i) + r\hat{X}(t_i)(t_{i+1} - t_i) + \sqrt{\sigma_0^2 + \sigma_1^2 \hat{X}^2(t_i)} \sqrt{t_{i+1} - t_i} Z_{i+1} \\
&\quad + \frac{1}{2} \left(\frac{2\sigma_1^2 \hat{X}(t_i)}{2\sqrt{\sigma_0^2 + \sigma_1^2 \hat{X}^2(t_i)}} \right) \left(\sqrt{\sigma_0^2 + \sigma_1^2 \hat{X}^2(t_i)} \right) (t_{i+1} - t_i) (Z_{i+1}^2 - 1) \\
&= \hat{X}(t_i) + r\hat{X}(t_i)(t_{i+1} - t_i) + \sqrt{\sigma_0^2 + \sigma_1^2 \hat{X}^2(t_i)} \sqrt{t_{i+1} - t_i} Z_{i+1} \\
&\quad + \frac{1}{2} \sigma_1^2 \hat{X}(t_i) (t_{i+1} - t_i) (Z_{i+1}^2 - 1).
\end{aligned}$$

Problem 3

(a)

$$F(x) = \int \frac{1}{\sigma \sqrt{x}} dx = \frac{2}{\sigma} \sqrt{x}, \quad Y(t) = F(X(t)) = \frac{2}{\sigma} \sqrt{X(t)}, \quad \text{and}$$

$$X(t) = \frac{\sigma^2 Y^2(t)}{4}.$$

By Itô's lemma,

$$\begin{aligned}
dY(t) &= \frac{\partial Y}{\partial X} dX(t) + \frac{1}{2} \frac{\partial^2 Y}{\partial X^2} (dX(t))^2 \\
&= \frac{2}{\sigma} \left(\frac{1}{2} \right) \frac{1}{\sqrt{X(t)}} \left[a(b - X(t)) dt + \sigma \sqrt{X(t)} dW(t) \right] \\
&\quad + \left(\frac{1}{2} \right) \left(-\frac{1}{2\sigma} \right) \frac{1}{\sqrt{X^3(t)}} (\sigma^2 X(t) dt) \\
&= a \left(\frac{b}{\sigma \sqrt{X(t)}} - \frac{\sqrt{X(t)}}{\sigma} - \frac{\sigma}{4a \sqrt{X(t)}} \right) dt + dW(t).
\end{aligned}$$

(b)

$$F(x) = \int \frac{1}{\sigma x} dx = \frac{1}{\sigma} \ln x, \quad Y(t) = F(X(t)) = \frac{1}{\sigma} \ln X(t), \quad \text{and}$$

$$X(t) = e^{\sigma Y(t)}.$$

By Itô's lemma,

$$\begin{aligned}
dY(t) &= \frac{\partial Y}{\partial X} dX(t) + \frac{1}{2} \frac{\partial^2 Y}{\partial X^2} (dX(t))^2 \\
&= \frac{1}{\sigma X(t)} \left[(r_0 + r_1 X(t)) dt + \sigma X(t) dW(t) \right] \\
&\quad - \frac{1}{2\sigma X^2(t)} (\sigma^2 X^2(t) dt) \\
&= \left(\frac{r_0}{\sigma X(t)} - \frac{r_1}{\sigma} - \frac{\sigma}{2} \right) dt + dW(t).
\end{aligned}$$

Problem 4

Let $Y = e^{-rT} [S(T) - K]^+$.

with

$$S(T) = S(0) e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z}, \quad Z \sim N(0,1),$$

and take θ to be σ , with $r, S(0), T$ and K positive constants.

By the method of pathwise differentiation, the vega (ν) of a standard European call option with maturity T and strike price K at time 0 is

$$\nu = E \left[\frac{dY}{d\sigma} \right].$$

Applying the chain rule for differentiation, we get

$$\frac{dY}{d\sigma} = \frac{dY}{dS(T)} \frac{dS(T)}{d\sigma}.$$

For the first of these two factors, we get

$$\frac{dY}{dS(T)} = e^{-rT} \mathbf{1}_{\{S(T) > K\}}.$$

For the second factor, we have

$$\frac{dS(T)}{d\sigma} = S(0) e^{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z} \left(-\sigma T + \sqrt{T}Z\right).$$

Eliminating Z , we have

$$\frac{dS(T)}{d\sigma} = S(T) \left[\frac{1}{\sigma} \ln \frac{S(T)}{S(0)} - \left(\frac{r}{\sigma} + \frac{\sigma}{2} \right) T \right].$$

Hence,

$$v = E \left[\mathbf{1}_{\{S(T) > K\}} e^{-rT} S(T) \left(\frac{1}{\sigma} \ln \left(\frac{S(T)}{S(0)} \right) - \left(\frac{r}{\sigma} + \frac{\sigma}{2} \right) T \right) \right].$$

Therefore,

$$f(S(0), S(T)) = S(T) \left(\frac{1}{\sigma} \ln \left(\frac{S(T)}{S(0)} \right) - \left(\frac{r}{\sigma} + \frac{\sigma}{2} \right) T \right).$$

Problem 5

The payoff of the given Asian option is a function of $S(t_1), \dots, S(t_m)$.

Under GBM,

$$S(t_i) = S(t_{i-1}) \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) h + \sigma \sqrt{h} Z_i \right), \quad i = 1, \dots, m,$$

where $Z_1, \dots, Z_m$ are i.i.d. standard normal random variables.

Let $g(x_1, x_2, \dots, x_m)$ be the joint pdf of $S(t_1), \dots, S(t_m)$.

Using the Markov property of GBM, we have

$$g(x_1, \dots, x_m) = g_1(x_1 | x_0) g_2(x_2 | x_1) \cdots g_m(x_m | x_{m-1}), \quad x_0 = S(0),$$

where each $g_j(x_j | x_{j-1}), j = 1, \dots, m$, is the transition density from time t_{j-1} to time t_j ,

$$g_j(x_j | x_{j-1}) = \frac{1}{x_j \sigma \sqrt{h}} f\left(\xi(x_j | x_{j-1})\right)$$

with

$\xi(x_j | x_{j-1}) = \frac{\ln\left(\frac{x_j}{x_{j-1}}\right) - \left(r - \frac{1}{2}\sigma^2\right)h}{\sigma\sqrt{h}}$ and $f(x)$ is the pdf of the standard normal distribution.

The score is thus given by

$$\frac{\partial \ln g(S(t_1), \dots, S(t_m))}{\partial \sigma} = \sum_{j=1}^m \frac{\partial \ln g_j(S(t_j) | S(t_{j-1}))}{\partial \sigma}.$$

For each term in $\frac{\partial \ln g(S(t_1), \dots, S(t_m))}{\partial \sigma}$, we have

$$\begin{aligned} \frac{\partial \ln g_j(S(t_j) | S(t_{j-1}))}{\partial \sigma} &= \frac{\partial}{\partial \sigma} \left[\ln\left(\frac{1}{S(t_j)\sqrt{h}}\right) - \ln \sigma + \ln \frac{1}{\sqrt{2\pi}} - \frac{\xi^2(S(t_j) | S(t_{j-1}))}{2} \right] \\ &= -\frac{1}{\sigma} - \xi(S(t_j) | S(t_{j-1})) \frac{\partial \xi(S(t_j) | S(t_{j-1}))}{\partial \sigma}. \end{aligned}$$

Since

$$\begin{aligned} \frac{\partial \xi(S(t_j) | S(t_{j-1}))}{\partial \sigma} &= \frac{\ln\left(\frac{S(t_j)}{S(t_{j-1})}\right) - rh}{\sqrt{h}} \left(-\frac{1}{\sigma^2} \right) + \frac{1}{2}\sqrt{h} \\ &= \frac{\left(\sigma\sqrt{h}Z_j - \frac{1}{2}\sigma^2h \right) \left(-\frac{1}{\sigma^2} \right)}{\sqrt{h}} + \frac{1}{2}\sqrt{h} \\ &= -\frac{1}{\sigma}Z_j + \sqrt{h}, \end{aligned}$$

we have

$$\frac{\partial \ln g_j(S(t_j) | S(t_{j-1}))}{\partial \sigma} = -\frac{1}{\sigma} - Z_j \left[-\frac{1}{\sigma}Z_j + \sqrt{h} \right] = \frac{Z_j^2 - 1}{\sigma} - \sqrt{h}Z_j.$$

Hence,

$$\frac{\partial \ln g(S(t_1), \dots, S(t_m))}{\partial \sigma} = \sum_{j=1}^m \left(\frac{Z_j^2 - 1}{\sigma} - \sqrt{h} Z_j \right).$$

The LRM estimator is

$$e^{-rT} (\bar{S}_A - K)^+ \sum_{j=1}^m \left(\frac{Z_j^2 - 1}{\sigma} - \sqrt{h} Z_j \right).$$

Problem 6

Let $S(t)$ be the stock price at time t .

From the given sample paths of the stock price process, terminal payoffs corresponding to each path, Y_3 , are given by

Path	$S(1.5)$	$Y_3 = \max(20 - S(1.5), 0)$
1	17.16	2.84
2	25.89	0
3	37.94	0
4	10.67	9.33
5	11.05	8.95
6	34.57	0

At $t = 1$, we model the expected payoff from continuation as a linear function, $f_2(S(1))$, of $S(1)$. Coefficients of the $f_2(S(1))$ are estimated from the in-the-money data in the following table by the least squares method.

Path	$Y_3 e^{-0.06 \times 0.5}$	$S(1)$	Exercise in-the-money?
1	2.76	22.82	No
2	0	15.84	Yes
3	0	38.63	No
4	9.05	13.26	Yes
5	8.69	14.68	Yes
6	0	22.78	No

Applying the regression function in the scientific calculator to the in-the-money data, we have

$$E \left[Y_3 e^{-0.06 \times 0.5} \mid S(1) \right] = -3.3878 \times S(1) + 55.353 := f_2(S(1)).$$

With this conditional expectation function, $f_2(S(1))$, the payoffs, Y_2 , at $t = 1$ is given by

$$Y_2 = \begin{cases} 20 - S(1), & \text{if } 20 - S(1) \geq f_2(S(1)), \\ e^{-0.03} Y_3, & \text{otherwise.} \end{cases}$$

The following table gives Y_2 for each sample path.

Path	Exercise $20 - S(1)$	Continuation $f_2(S(1))$	$Y_3 e^{-0.03}$	Y_2
1	0	0	2.76	2.76
2	4.16	1.69	0	4.16
3	0	0	0	0
4	6.74	10.43	9.05	9.05
5	5.32	5.62	8.69	8.69
6	0	0	0	0

Next, we repeat the procedure for $t = 0.5$ for the data in the following table,

Path	$Y_2 e^{-0.03}$	$S(0.5)$	Exercise in-the-money?
1	2.68	21.74	No
2	4.04	17.98	Yes
3	0	28.07	No
4	8.78	10.38	Yes
5	8.43	21.98	No
6	0	18.88	Yes

we have

$$E[Y_2 e^{-0.03} | S(0.5)] = -0.8736 \times S(0.5) + 18.0303 := f_1(S(0.5)).$$

With this conditional expectation function, $f_1(S(0.5))$, the payoffs, Y_1 , at $t = 0.5$ is given by

$$Y_1 = \begin{cases} 20 - S(0.5), & \text{if } 20 - S(0.5) \geq f_1(S(0.5)), \\ e^{-0.03} Y_2, & \text{otherwise.} \end{cases}$$

The following table gives Y_1 for each sample path.

Path	Exercise $20 - S(0.5)$	Continuation $f_1(S(0.5))$	$Y_2 e^{-0.03}$	Y_1
1	0	0	2.68	2.68
2	2.02	2.32	4.04	4.04
3	0	0	0	0
4	9.62	8.96	8.78	9.62
5	0	0	8.43	8.43
6	1.12	1.54	0	0

Finally, the price of the American option at $t = 0$ is estimated by the average of $e^{-0.03} Y_1$, that is, \$4.0063.

Problem 7

(a)

The value of the portfolio is

$$V(t, S_1, S_2) = 2C(S_1) - 3P(S_2) - 4S_2 + 5S_1.$$

The delta-gamma approximation of ΔV is given by

$$\begin{aligned} \Delta V &\approx \frac{\partial V}{\partial t} \Delta t + \frac{\partial V}{\partial S_1} \Delta S_1 + \frac{\partial V}{\partial S_2} \Delta S_2 + \frac{1}{2} \frac{\partial^2 V}{\partial S_1^2} (\Delta S_1)^2 + \frac{\partial^2 V}{\partial S_1 \partial S_2} (\Delta S_1)(\Delta S_2) + \frac{1}{2} \frac{\partial^2 V}{\partial S_2^2} (\Delta S_2)^2 \\ &= \frac{\partial V}{\partial S_1} \Delta S_1 + \frac{\partial V}{\partial S_2} \Delta S_2 + \frac{1}{2} \frac{\partial^2 V}{\partial S_1^2} (\Delta S_1)^2 + \frac{\partial^2 V}{\partial S_1 \partial S_2} (\Delta S_1)(\Delta S_2) + \frac{1}{2} \frac{\partial^2 V}{\partial S_2^2} (\Delta S_2)^2. \end{aligned}$$

Since $L = -\Delta V$, let Q be the delta-gamma approximation of L ,

$$\begin{aligned} Q &= -\frac{\partial V}{\partial S_1} \Delta S_1 - \frac{\partial V}{\partial S_2} \Delta S_2 - \frac{1}{2} \frac{\partial^2 V}{\partial S_1^2} (\Delta S_1)^2 - \frac{\partial^2 V}{\partial S_1 \partial S_2} (\Delta S_1)(\Delta S_2) - \frac{1}{2} \frac{\partial^2 V}{\partial S_2^2} (\Delta S_2)^2 \\ &= -\left(2 \frac{\partial C}{\partial S_1} + 5\right) \Delta S_1 - \left(-3 \frac{\partial P}{\partial S_2} - 4\right) \Delta S_2 - \frac{1}{2} \left(2 \frac{\partial^2 C}{\partial S_1^2}\right) (\Delta S_1)^2 - \frac{1}{2} \left(-3 \frac{\partial^2 P}{\partial S_2^2}\right) (\Delta S_2)^2 \\ &= -6.12 \Delta S_1 + 3.19 \Delta S_2 - 0.08 (\Delta S_1)^2 + 0.105 (\Delta S_2)^2 \\ &= \alpha^T \Delta S + \Delta S^T \Gamma \Delta S, \end{aligned}$$

where

$$\alpha^T = (-6.12, 3.19)^T \quad \text{and} \quad \Gamma = \begin{pmatrix} -0.08 & 0 \\ 0 & 0.105 \end{pmatrix}.$$

(b)

Let

$$\tilde{C} = \begin{pmatrix} \tilde{C}_{11} & 0 \\ \tilde{C}_{12} & \tilde{C}_{22} \end{pmatrix}$$

be the Cholesky factor of Σ .

We have

$$\begin{pmatrix} \tilde{C}_{11} & 0 \\ \tilde{C}_{12} & \tilde{C}_{22} \end{pmatrix} \begin{pmatrix} \tilde{C}_{11} & \tilde{C}_{12} \\ 0 & \tilde{C}_{22} \end{pmatrix} = \begin{pmatrix} 0.5 & 0.18 \\ 0.18 & 0.3 \end{pmatrix}$$

$$(\tilde{C}_{11})^2 = 0.5 \Rightarrow \tilde{C}_{11} = \sqrt{0.5} = 0.7071,$$

$$\tilde{C}_{12}\tilde{C}_{11} = 0.18 \Rightarrow \tilde{C}_{12} = 0.2546,$$

$$(\tilde{C}_{12})^2 + (\tilde{C}_{22})^2 = 0.3 \Rightarrow \tilde{C}_{22} = 0.485.$$

So,

$$\begin{aligned} A &= \tilde{C}^T \Gamma \tilde{C} = \begin{pmatrix} 0.7071 & 0.2546 \\ 0 & 0.485 \end{pmatrix} \begin{pmatrix} -0.08 & 0 \\ 0 & 0.105 \end{pmatrix} \begin{pmatrix} 0.7071 & 0 \\ 0.2546 & 0.485 \end{pmatrix} \\ &= \begin{pmatrix} -0.0566 & 0.0267 \\ 0 & 0.0509 \end{pmatrix} \begin{pmatrix} 0.7071 & 0 \\ 0.2546 & 0.485 \end{pmatrix} \\ &= \begin{pmatrix} -0.0332 & 0.013 \\ 0.013 & 0.0247 \end{pmatrix}. \end{aligned}$$

Let λ be the eigenvalue of A . From the hint, we have $\lambda_1 = -0.0360$ and $\lambda_2 = 0.0275$.

Also, the eigenvector belonging to $\lambda_1 = -0.0360$ is

$$\mathbf{v}_1 = \begin{pmatrix} 0.9778 \\ -0.2095 \end{pmatrix},$$

and the eigenvector belonging to $\lambda_2 = 0.0275$ is

$$\mathbf{v}_2 = \begin{pmatrix} 0.2095 \\ 0.9778 \end{pmatrix}.$$

Let

$$\Lambda = \begin{pmatrix} -0.036 & 0 \\ 0 & 0.0424 \end{pmatrix} \text{ and } U = (\mathbf{v}_1 \quad \mathbf{v}_2) = \begin{pmatrix} 0.9778 & 0.2095 \\ -0.2095 & 0.9778 \end{pmatrix}.$$

Setting

$$C = \tilde{C}U = \begin{pmatrix} 0.7071 & 0 \\ 0.2546 & 0.485 \end{pmatrix} \begin{pmatrix} 0.9778 & 0.2095 \\ -0.2095 & 0.9778 \end{pmatrix} = \begin{pmatrix} 0.6914 & 0.1481 \\ 0.1473 & 0.5276 \end{pmatrix}.$$

Let $\Delta S = CZ$.

So,

$$\begin{aligned} Q &= \alpha^T CZ + Z^T (C^T \Gamma C) Z \\ &= -3.7614Z_1 + 0.7764Z_2 - 0.036Z_1^2 + 0.0275Z_2^2. \end{aligned}$$

Problem 8

Consider the moment generating function of Z under P_θ ,

$$\begin{aligned} &E_\theta \left[e^{\beta^T Z} \right] \\ &= E \left[e^{\theta Q - \psi(\theta) + \beta^T Z} \right] \\ &= E \left[\exp \left(a\theta + \theta \sum_{j=1}^m (b_j Z_j + \lambda_j Z_j^2) - a\theta - \frac{1}{2} \sum_{j=1}^m \left(\frac{\theta^2 b_j^2}{1 - 2\theta \lambda_j} - \log(1 - 2\theta \lambda_j) \right) + \sum_{j=1}^m \beta_j Z_j \right) \right] \\ &= \exp \left(-\frac{1}{2} \sum_{j=1}^m \left(\frac{\theta^2 b_j^2}{1 - 2\theta \lambda_j} - \log(1 - 2\theta \lambda_j) \right) \right) E \left[\exp \left(\theta \sum_{j=1}^m (b_j Z_j + \lambda_j Z_j^2) + \sum_{j=1}^m \beta_j Z_j \right) \right] \\ &= \exp \left(-\frac{1}{2} \sum_{j=1}^m \left(\frac{\theta^2 b_j^2}{1 - 2\theta \lambda_j} - \log(1 - 2\theta \lambda_j) \right) \right) \exp \left(\frac{1}{2} \sum_{j=1}^m \left(\frac{(\theta b_j + \beta_j)^2}{1 - 2\theta \lambda_j} - \log(1 - 2\theta \lambda_j) \right) \right) \\ &= \exp \left(\sum_{j=1}^m \frac{\theta b_j \beta_j}{1 - 2\theta \lambda_j} + \frac{1}{2} \sum_{j=1}^m \frac{\beta_j^2}{1 - 2\theta \lambda_j} \right). \end{aligned}$$

Hence, the distribution of Z under P_θ is a multivariate normal

$$Z \sim N(\mu(\theta), \Sigma(\theta)),$$

where $\Sigma(\theta)$ is a diagonal matrix with diagonal entries $\sigma_j^2(\theta)$,

$$\mu_j(\theta) = \frac{\theta b_j}{1 - 2\lambda_j \theta}, \quad \sigma_j^2(\theta) = \frac{1}{1 - 2\lambda_j \theta}.$$

Problem 9

(a)

$$\begin{aligned} L(\sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X_i - \mu)^2}{2\sigma^2}} \\ &\propto \prod_{i=1}^n \frac{1}{\sigma} e^{-\frac{(X_i - \mu)^2}{2\sigma^2}} \\ &= (\sigma^2)^{-\frac{n}{2}} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2\right]. \end{aligned}$$

(b)

Conditional distribution

$$\begin{aligned} \pi(\sigma^2 | x) &\propto L(\sigma^2) p(\sigma^2) \\ &\propto (\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2\right\} \frac{\beta^\alpha}{\Gamma(\alpha)} (\sigma^2)^{-\alpha-1} e^{-\frac{\beta}{\sigma^2}} \\ &\propto (\sigma^2)^{-\frac{n}{2}-\alpha-1} \exp\left[-\frac{1}{\sigma^2} \left[\frac{1}{2} \sum_{i=1}^n (X_i - \mu)^2 + \beta\right]\right]. \end{aligned}$$

The posterior is $IG\left(\alpha + \frac{n}{2}, \beta + \frac{1}{2} \sum_{i=1}^n (X_i - \mu)^2\right)$.

Problem 10

Let

$$Q = (q_{ij})_{i,j=1,\dots,4}$$

be the given transition matrix.

Since our target limiting distribution is $(1/6, 1/6, 1/3, 1/3)$, so

$$\pi(1) = 1/6, \pi(2) = 1/6, \pi(3) = 1/3, \pi(4) = 1/3.$$

For $i \neq j$ and $q_{ij} \neq 0$, evaluating

$$\alpha_{ij} = \min \left[1, \frac{\pi(j)q_{ji}}{\pi(i)q_{ij}} \right],$$

we have

$$\begin{aligned} \alpha_{13} &= \min \left[1, \frac{\pi(3)q_{31}}{\pi(1)q_{13}} \right] = \min \left[1, \frac{(1/3)0}{(1/6)(1/2)} \right] = 0, & \alpha_{14} &= \min \left[1, \frac{\pi(4)q_{41}}{\pi(1)q_{14}} \right] = \min \left[1, \frac{(1/3)(1/4)}{(1/6)(1/3)} \right] = 1, \\ \alpha_{23} &= \min \left[1, \frac{\pi(3)q_{32}}{\pi(2)q_{23}} \right] = \min \left[1, \frac{(1/3)(1/2)}{(1/6)(1/3)} \right] = 1, & \alpha_{24} &= \min \left[1, \frac{\pi(4)q_{42}}{\pi(2)q_{24}} \right] = \min \left[1, \frac{(1/3)(1/4)}{(1/6)(1/3)} \right] = 1, \\ \alpha_{32} &= \min \left[1, \frac{\pi(2)q_{23}}{\pi(3)q_{32}} \right] = \min \left[1, \frac{(1/6)(1/3)}{(1/3)(1/2)} \right] = \frac{1}{3}, & \alpha_{34} &= \min \left[1, \frac{\pi(4)q_{43}}{\pi(3)q_{34}} \right] = \min \left[1, \frac{(1/3)(1/4)}{(1/3)(1/2)} \right] = \frac{1}{2}, \\ \alpha_{41} &= \min \left[1, \frac{\pi(1)q_{14}}{\pi(4)q_{41}} \right] = \min \left[1, \frac{(1/6)(1/3)}{(1/3)(1/4)} \right] = \frac{2}{3}, & \alpha_{42} &= \min \left[1, \frac{\pi(2)q_{24}}{\pi(4)q_{42}} \right] = \min \left[1, \frac{(1/6)(1/3)}{(1/3)(1/4)} \right] = \frac{2}{3}, \\ \alpha_{43} &= \min \left[1, \frac{\pi(3)q_{34}}{\pi(4)q_{43}} \right] = \min \left[1, \frac{(1/3)(1/2)}{(1/3)(1/4)} \right] = 1. \end{aligned}$$

In summary, we have

$\alpha_{i,j}$	1	2	3	4
1	--	--	0	1
2	--	--	1	1
3	--	1/3	--	1/2
4	2/3	2/3	1	--

The revised transition probabilities are given as follows

$$p_{ij} = \alpha_{ij}q_{ij}, \text{ for all } i \neq j \text{ and } p_{ii} = 1 - \sum_{j \neq i} p_{ij}.$$

$$P = \begin{pmatrix} 2/3 & 0 & 0 & 1/3 \\ 0 & 1/3 & 1/3 & 1/3 \\ 0 & 1/6 & 7/12 & 1/4 \\ 1/6 & 1/6 & 1/4 & 5/12 \end{pmatrix}.$$

The transition matrix can be verified by checking the equation $\pi P = \pi$.

Problem 11

If x_{k-1} is the current point and x_k is the a prospective variate then the α in the Metropolis-Hastings algorithm is

$$\begin{aligned} \alpha(x_{k-1}, x_k) &= \frac{f(x_k)q(x_{k-1} | x_k)}{f(x_{k-1})q(x_k | x_{k-1})} \\ &= \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{x_k^2}{2}} \frac{1}{2} e^{-|x_{k-1}|}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{x_{k-1}^2}{2}} \frac{1}{2} e^{-|x_k|}} \\ &= \frac{e^{-\frac{x_k^2}{2}} \cdot e^{-|x_{k-1}|}}{e^{-\frac{x_{k-1}^2}{2}} \cdot e^{-|x_k|}}. \end{aligned}$$

The algorithm starts with $X_0 = 0$.

k	1	2	3	4
(U, V)	(0.18, 0.87)	(0.02, 0.42)	(0.31, 0.08)	(0.12, 0.79)
X_k	$V = 0.87 > 0.5,$ $X_k = -\ln 0.18$ $= 1.7148.$	$V = 0.42 < 0.5,$ $X_k = \ln 0.02$ $= -3.912.$	$V = 0.08 < 0.5,$ $X_k = \ln 0.31$ $= -1.1712.$	$V = 0.79 > 0.5,$ $X_k = -\ln 0.12$ $= 2.1203.$
α	1.277	0.0186	1.2723	0.5418
W	0.55	0.03	0.18	0.43
θ_k	$\alpha > 1,$ $\theta_k = X_k = 1.7148$	$W = 0.03 > \alpha,$ $\theta_k = \theta_{k-1} = 1.7148$	$\alpha > 1,$ $\theta_k = X_k = -1.1712$	$W = 0.43 < \alpha,$ $\theta_k = X_k = 2.1203$
X_k	1.7148	$X_k = X_{k-1} = 1.7148$	-1.1712	2.1203
Remark	State changes	State unchanged	State changes	State changes

The detailed calculation of α :

$k = 1$

$$\alpha = \frac{e^{-0} \cdot e^{-\frac{(1.7148)^2}{2}}}{e^{-|1.7148|} \cdot e^{-\frac{0}{2}}} = 1.277.$$

$k = 2$

$$\alpha = \frac{e^{-|1.7148|} \cdot e^{-\frac{(-3.912)^2}{2}}}{e^{-|-3.912|} \cdot e^{-\frac{(1.7148)^2}{2}}} = 0.0186.$$

$k = 3$

$$\alpha = \frac{e^{-|1.7148|} \cdot e^{-\frac{(-1.1712)^2}{2}}}{e^{-|-1.1712|} \cdot e^{-\frac{(1.7148)^2}{2}}} = 1.2723.$$

$$k = 4$$

$$\alpha = \frac{e^{-|-1.1712|} \cdot e^{-\frac{(2.1203)^2}{2}}}{e^{-|2.1203|} \cdot e^{-\frac{(-1.1712)^2}{2}}} = 0.5418.$$